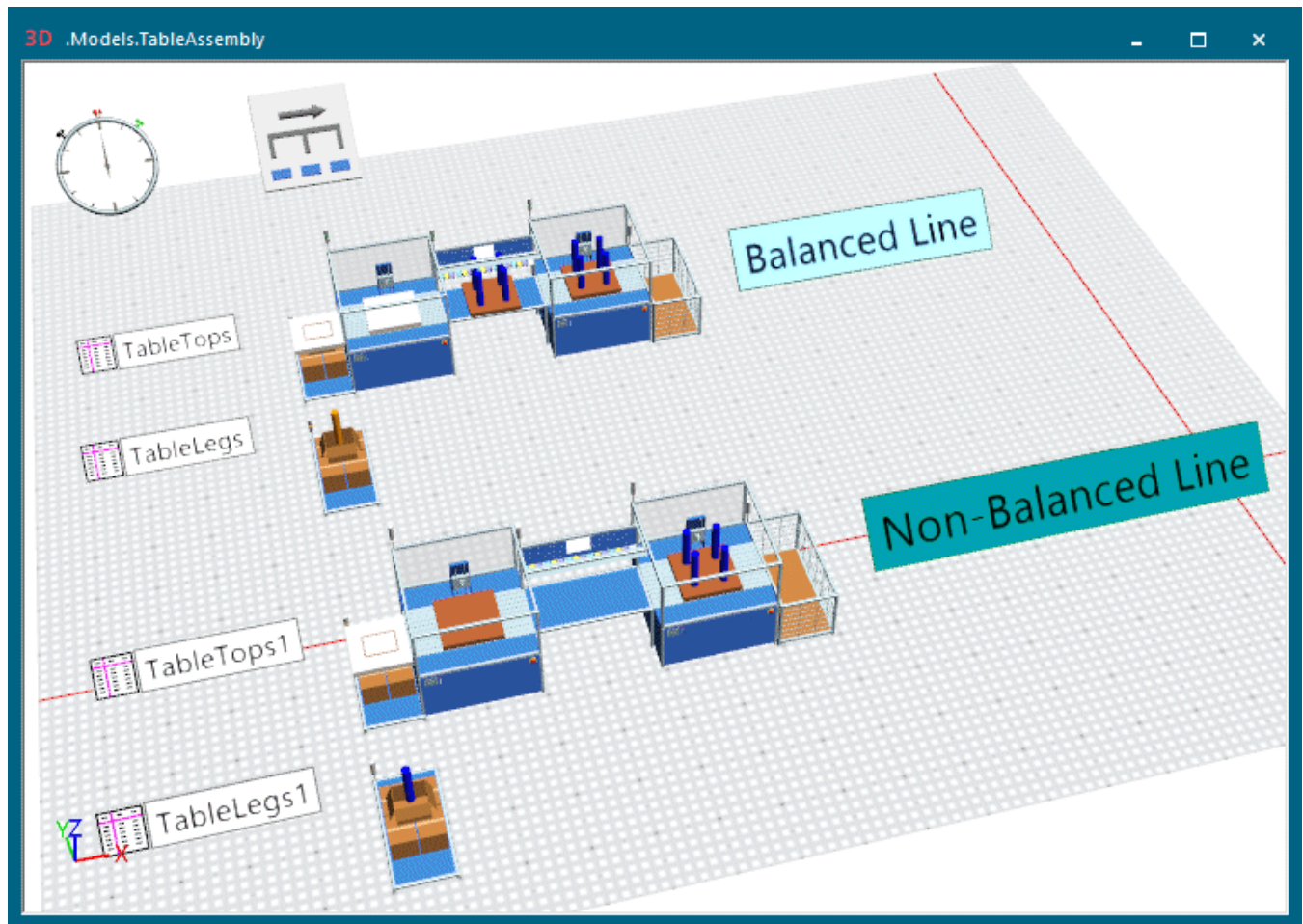




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Modeling Random Processes

Model Random Processes

Many computer programs use random number generators to create a stream of random numbers that generally are located in the interval between 0 and 1. Starting from given start values, the so-called seed values, a large number of stochastically independent random numbers has to be available.

Compare these examples of the application of random numbers in computer programs:

- Gaming programs use random numbers to allow for a greater diversity of the operating sequence of the game.
- Database programs use random numbers to enter a large number of data sets and thus test the functionality of the database.
- Simulation programs, such as *Plant Simulation*, use random numbers to represent, among others, the interval between failures of machines.

Compare [Simulating Random Processes](#)

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Random Numbers and Their Statistical Distribution

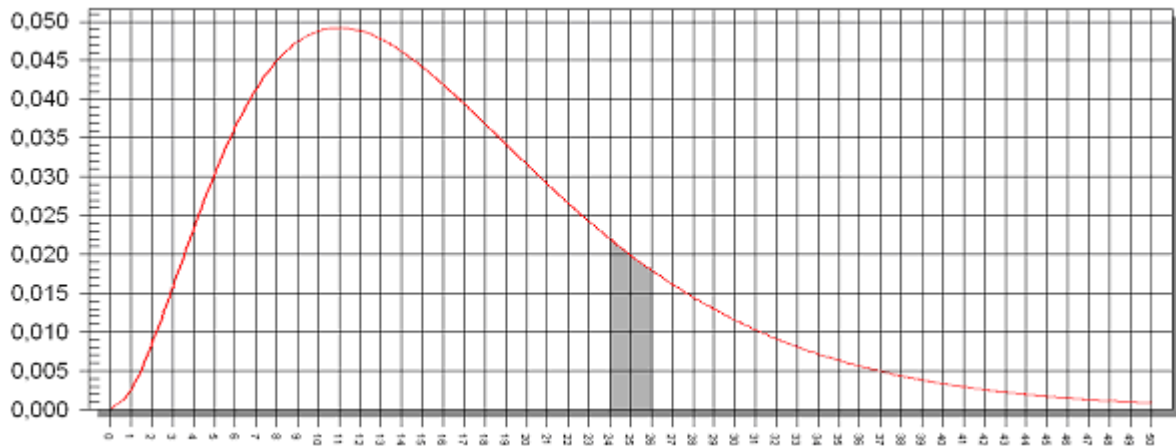
We know that there are deterministic processes and random processes. For a deterministic process, we can predict the result of the process, for example by knowing a law of nature. If we do not know all factors that characterize a process, its result is random.

When we simulate production processes, the occurrence of pauses is a deterministic process. Failures of machines or the rejects that are produced, on the other hand, are random processes. A random process results in a random number. The replication of the process results in the realization of the random number. Throwing the dice is a random process, the resulting number of points is the associated random number.

There are random numbers that can have individual, separate values. These may be a limited or an unlimited number of different values, such as the number of rejects over a long period of time or the number of orders during a day. A limited number may result from throwing the dice, an unlimited number can be a countable numbered amount. These types of random numbers are called **discrete**. To simplify matters, we suppose that these values can be 0, 1, 2, etc. For describing the random number we specify the probability for the individual numbers to be the result of throwing the dice. Adding up all probabilities has to result in 1, as one of the numbers will always be thrown.

A random number that can take all values within a limited or an unlimited interval of numbers is **continuous**. Examples of continuous random numbers are the mean time between failures (**MTBF**) and the mean time to repair (**MTTR**) of a machine. For describing a continuous random number, we cannot specify the probability for an individual number to occur. We rather have to specify the probability with which this random number is located between two given values, i.e., we specify an interval.

The **probability density function** test describes the distribution of a random number. The density function of a continuous random number only has values greater than 0 or equal to 0. It can be continuously drawn without interruption. The area underneath the function between two values a and b is the probability that the random number is located between these values a and b. The value of the density function at the position x describes how often a value will occur that approximates x. The entire area below the curve of the density function has the value 1, as with each realization of the random number a certain value occurs.



The screenshot above shows the density function of the gamma distribution with the parameters Alpha = 3 and Beta = 5.5. The most random numbers occur in the vicinity of the apex of the density function (random number $x = 11$). This value is called the **modal value**. When you build the mean of many realizations of this distribution, you will notice that the mean value is substantially greater than the modal value. The mean value will be around 16.5. This gamma distribution will create random numbers between 24 and 26 with a probability of 0.04. This probability equals the gray area below the curve in the screenshot above.

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Use Pseudo Random Numbers

On first sight it seems to be impossible to simulate random processes with a computer, as a computer computes the numbers according to fixed stipulated calculation rules. A computer cannot create a truly random sequence of numbers. A computer-generated sequence of numbers can only approximately have the properties of random numbers. This is why they are called **pseudo random numbers**.

The sequence of numbers that a computer creates is called **random number stream**. The computer computes the following number starting from a random number. The algorithm for creating random numbers starts out with the so-called **seed values** and can generate any number of values from them. By using different seed values you can model several random processes that are independent of each other.

For creating a random number it oftentimes suffices to create a uniformly distributed random number located in the interval between 0 and 1 (0,1). A random number is uniformly distributed, when the probability for an interval only depends on its length, and not from its position on the line of numbers. Using a uniformly distributed random number the computer can, for example, algorithmically create a normally distributed random number with a given mean value and standard deviation.

A uniformly distributed pseudo random number has to meet a number of requirements:

- The arrangement of the random numbers shows no typical characteristics, i.e., the algebraic signs of consecutive random numbers do not display any typical patterns.

- The algorithm for creating random numbers has to become periodic, as the computer only has a finite number of states. For this reason the period length has to be as large as possible.
- Starting from the given seed values a large number of stochastically independent random numbers has to be available for the simulation.
- The random processes of a simulation have to be reproducible to be able to use certain statistics methods, such as the variance reduction.

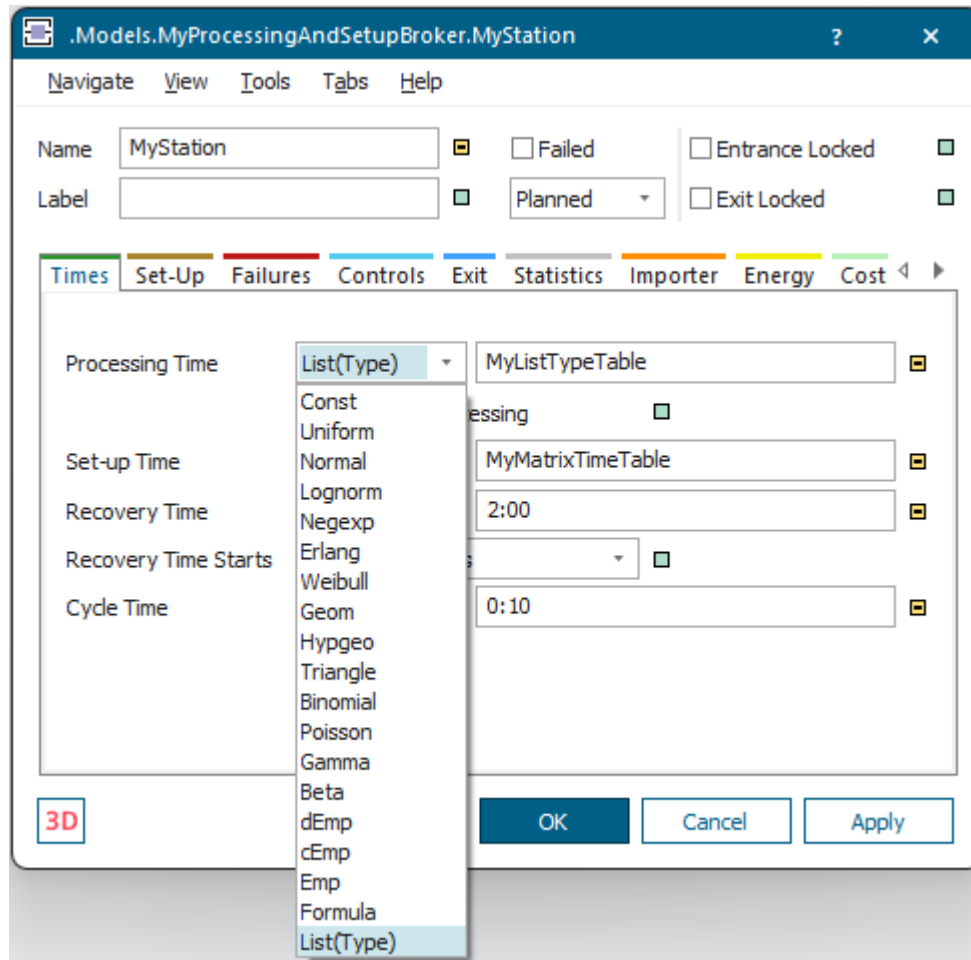
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Use Probability Distributions

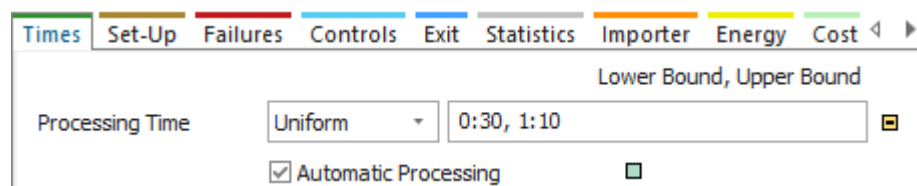
As a rule, only very few observed data about random process, such as the interval between two failures of a machine, will be available to you.

To allow you to reproduce these random processes in your simulation model, *Plant Simulation* provides a number of probability distributions, compare [Probability Distributions](#), [Empirical Distributions](#), and [User-defined Distributions](#).

Step 3: Decide Which Distribution to Use assists you in picking the right distribution for your purpose with the object *DataFit*.



Once you select the distribution from the list, *Plant Simulation* shows the parameters, which this distribution requires, along the upper border of the tab.



Calculate the respective values in the observed data, which you received from your customer, and type them into the text box. The lower bound and the upper bound are optional, meaning that you can, but do not have to specify them.

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